The Impact of Instagram Followers, Age, Brand Endorsements, and Matches Played on Professional Football Players' Market Value

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Abstract

Background/Objective

In today's day and age, professional footballers are no longer just meant to be valued for their performance on the field but also for their off the field attributes such as their age, experience, social media presence, and marketability. Social media, specifically Instagram, has turned into a valuable asset to develop a player's profile and brand; age and number of matches played indicates experience, development, potential and longevity of a player's career. However, how many Instagram followers, age, number of brand endorsement industries and number of matches played are associated with a professional football player's market value are still not known. This study analyzes if any relationship exists between these variables and market value, as well as the magnitude of the relationship.

Methods

A multiple linear regression model was estimated in STATA MP 18 for this study. The data informed here includes player data (name, age, nationalities, market value). Data from the dataset was complemented with the number of Instagram followers, number of brand-endorsement industries, and number of matches played of the players. The dependent variable was the market value, and the independent variables were the number of countries the player is from, the number of Instagram followers, number of brand-endorsement industries, number of matches played and age of the player. Prior to running the regression, scatter plots were created for each predictor against the market value to look for linearity.

Results

The regression model was statistically significant, $F(5, 475) = 311.15$, $p < .001$, and explained approximately 76.6% of the variance in market value ($R^2 = .7661$, Adjusted $R^2 = .7636$). Market value was positively predicted by number of matches played ($\beta = 0.822$, $p < .001$), number of Instagram followers ($\beta = 0.0006$, $p < .001$), and the country development category, where a player was from developed/developing/and underdeveloped region ($\beta = 10.01$, $p < .001$). The number of brand-endorsement industries approached significance ($\beta = 18.06$, $p = .054$) and age was not a significant predictor ($\beta = -0.29$, $p = .123$). These trends were further confirmed as market value was strongly correlated with Instagram followers ($r = .769$, $p < .001$) and matches played ($r = .749$, $p < .001$), and moderately correlated with brand industries ($r = .565$, $p < .001$). Multicollinearity was not a serious issue (mean VIF = 2.59), but heteroskedasticity ($\chi^2(1) = 212.91$, $p < .001$) and possibly model misspecification (Ramsey RESET $F(3, 472) = 25.19$, $p < .001$) were identified from the diagnostic tests.

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Conclusions

The findings indicate that a player's market valuation is well-supported by experience in playing matches and commercial and social media presence as evidenced by the number of his Instagram followers and the number of brands he is associated with, but not by his age alone. This means that in today's game visibility and marketability can be more important than experience in influencing player-value by either clubs or the market. The model is explanatory, but with possible heteroskedasticity and omitted variable bias, it would benefit from being improved before the results are considered as definitive, for example by transforming variables or introducing on-pitch performance variables.

Introduction

Background and Context

One of the most popular sports in the world, and one of the biggest sporting industries, professional football is more important than ever to be able to understand and evaluate the factors that help define a player's market value. A total of 78,742 international player transfers took place all over the world in 2024, a sum of $8.59 billion spent by professional football clubs on international transfer fees (“International Transfers Reach Historic High In 2024”). Until this century one of the most important factors influencing market value has been age. Younger players may be worth a lot more than older players and have more potential for growth and years left in their careers.

But in recent years, the expansion and emergence of social media, particularly Instagram, has come as a new factor which has the potential to affect a footballer's market value in the last 10 years. Instagram has already reached 2 billion monthly active users globally.In fact, Instagram has already surpassed 2 billion monthly users worldwide (Statista, 2025). This wide-spread makes it possible for players to develop their personal brand, win sponsorship offers, as well as connect with millions of fans across the globe, making it possible for them to exploit brand-new commercial possibilities that do not exist for expert players.

Problem Statement and Rationale:

The question is that of a professional footballer, there are many factors that play a unique and interactive role in determining the professional worth of a player, and it is difficult to conclude which factors seem to be the most important ones for determining the worth of a professional footballer. Apart from player performance on the pitch, age and matches played are among the most influential factors of market value traditionally believed to hold sway (Margareta & Malinda, 2022).

Literature Review

There is some converging evidence but not quite consistent evidence on drivers of market value from prior research. Goals have been found to be one of the key drivers of market value of a football player, and a strong social media presence has been found to magnify ‘goals effect’ on market value (Cai, 2024). The odds ratio has been decreasing with age, peaking at U19 and dropping to 2.7, then 1.8 at U21 and U23, respectively, implying that experience and performance are key factors to valuation as players mature, with younger players (quartile 4) having higher market value than older ones (quartile 1) at older ages (Born, 2022).

Other research specifically refers to the commercial aspect of value. One study showed that the more people are following a player on social media the higher a player's market value is, which suggests that it is

increasing the media attention that a player receives, and that Instagram is especially directly correlated with a player's market value as recorded by Transfermarkt (Frenger et al., 2019). Consideration of age effects has also been done directly: players between the ages of 21 and 30 are shown to have a higher market value than players younger than 21 or older than 30 (Sánchez et al., 2022); another examination of top five European leagues reveals that the market value of a player is steadily decreasing as they get older (Liu, n.d.).

It's not just about the physical attributes, however; contractual aspects are also important, too, as evidenced by how the length of a player's contract is correlated with market value (Prakosa & Prasetio, 2025). The literature collectively indicates that this is due to a combination of career stage effects associated with age, on-field skills and experience, contract, and more and more lately social media and commercial profile of the player. This study contributes to this literature by simultaneously testing age, matches played, Instagram followers, brand endorseability and country of origin (developed/developing and underdeveloped) together in one multiple regression model.

Methods

This study employed multiple regression models, which was constructed using the software of STATA MP 18. Data was gathered from Transfermarkt, a website containing each player's name, age, market value and nationality. This was sampled to get a sample of players. The first 100 football players, from a dataset of 500 football players. The players were in descending order of market value (Transfermarkt, n.d.). Additional data was found by searching for it : Instagram followers, number of different industries where the player has an endorsement, and number of matches played. The final data set was exported and analyzed in STATA MP 18.

The dependent variable (Y) was the market value (in millions of euros) of each player at the Transfermarkt website. Age, number of Instagram followers, number of brand endorsement industries, number of matches played, and whether the player comes from a developing country (categorized) were the independent variables (X). In order to see if the data were linear, scatterplots for each independent variable were created versus the market value, and a linear regression method was used if the scatter plot appeared linear. Then pairwise (Pearson) correlation was conducted between market value and each of the key predictors, and the complete multiple regression model was conducted. Various model diagnostics were performed to check the assumptions of the model such as multicollinearity (variance inflation factors), heteroskedasticity (Breusch–Pagan/Cook–Weisberg test), and omitted variables (Ramsey RESET test).

H_{01}: Country development status has no significant association with market value.

H_{11}: Country development status has a significant association with market value.

H_{02}: Number of Instagram followers has no significant association with market value.

H_{12}: Number of Instagram followers has a significant association with market value.

H_{03}: Number of brand endorsements has no significant association with market value.

H_{13}: Number of brand endorsements has a significant association with market value.

H_{04}: Number of matches played has no significant association with market value.

H_{14}: Number of matches played has a significant association with market value.

Results

Descriptive Statistics

Table 1 shows the descriptive statistics for the study variables, with the number of observations, the mean, the standard deviation, the minimum and the maximum. The mean value of the markets was €42.23 million (SD = 24.67) with a range between €23 million and €200 million. The mean player age was 24.24 years (SD = 2.99, range 17–33). On average, players played 20.71 games (SD = 11.32, range 0–60) and had 3,901 Instagram followers (SD = 15,750.02, range 0–205,357), with a highly right-skewed distribution representing a small number of players having a disproportionately large number of followers. The mean values for the brand-endorsement and developing-country indicator variables were 1.01 (with an SD of .11) and .19 (with an SD of .51), respectively, for the 481 players without missing values. Some other coded variables (H to M) were not used as predictors in the final regression model, but were kept in the original data extraction.

Variable	Obs	Mean	Std. Dev.	Min	Max
Age	500	24.244	2.990	17	33
Brand Industry (code)	481	1	0.111	1	2
Market Value	500	42.228	24.672	23	200
No. of Matches Played	500	20.708	11.323	0	60
No. of Instagram Followers	500	3,901.462	15,750.02	0	205,357
Developing Country	500	0.192	0.506	0	3

Note. Obs = number of non-missing observations; Std. Dev. = standard deviation. Values for Market Value are reported in millions of euros.

Correlation Analysis

Before the multiple regression, pairwise Pearson correlation was performed for each of the three important continuous/ordinal predictors and market value to investigate the direction and strength of the bivariate relationships.

There we identified a good and positive correlation between the number of Instagram followers and market value, $r = .769$, $p < .001$.

	Market Value	IG Followers
Market Value	1.0000	
IG Followers	0.7694 ($p < .001$)	1.0000

The correlation between number of brand-endorsement industries and market value was also positive, and moderately strong, $r = .565$, $p < .001$.

	Market Value	Brand Industries
Market Value	1.0000	
Brand Industries	0.5652 ($p < .001$)	1.0000

The correlation between number of matches played and market value was positive and strong, $r = .749$, $p < .001$.

	Market Value	Matches Played
Market Value	1.0000	
Matches Played	0.7485 (p < .001)	1.0000

Regression Analysis

A multiple regression model was used with market value as the dependent variable and age, brand-endorsement industries, matches played, Instagram followers, and developing-country origin as predictors. The overall model was statistically significant and explained a substantial proportion of the variance in market value.

Source	SS	df	MS
Model	229,975.705	5	45,995.141
Residual	70,216.756	475	147.825
Total	300,192.462	480	625.401

Number of obs = 481; F(5, 475) = 311.15; Prob > F = 0.0000; R-squared = 0.7661; Adj R-squared = 0.7636; Root MSE = 12.158.

Predictor	Coef.	βStd. Err.	t	P>|t|	95% CI
Age	-0.288	0.186	-1.55	0.123	[-0.655, 0.078]
Brand Industry Code	18.063	9.341	1.93	0.054	[-0.291, 36.417]
No. of Matches Played	0.822	0.068	12.06	0.000	[0.688, 0.956]
No. of Instagram Followers	0.0006	0.0001	8.27	0.000	[0.0005, 0.0008]
Country development status	10.009	1.400	7.15	0.000	[7.258, 12.759]
Constant	9.678	10.738	0.90	0.368	[-11.421, 30.777]

Based on the P value, we reject the null hypothesis for number of matches played, number of instagram followers, country development status at 5% significance level. The null hypothesis for brand industry code has been rejected at 10% significance level. The strongest and most reliable positive predictor of the market value was number of matches played (β = 0.822, p < .001): For each match played, the market value increased by approximately €0.82 million, while keeping other factors constant. The number of Instagram followers was also a significant positive predictor (= 0.0006, p < .001); while small, such a large order (up to more than 200,000 followers) means that large follower counts lead to a meaningful increase in the predicted value of the market. Compared to other players, playing during development had players with an average market value of approximately €10 million (= 10.009, p < .001), while taking no other variables into account. The coefficient was positive for the number of brand endorsement industries, but did not reach conventional significance (= 18.063, p = .054). The coefficient for age is negative (= -0.288), and this is not a statistically significant predictor of market value when all other factors are included in the model (p = .123), meaning that age's impact on value is primarily through its relationship with experience (matches played) and not directly with it.

Model Diagnostics

Variance inflation factors (VIF) were calculated to check for multicollinearity among predictors. All VIF values were well below the common threshold of 10 (and below 5), with a mean VIF of 2.59, indicating that multicollinearity was not a serious concern in this model.

The Breusch–Pagan/Cook–Weisberg test was used to test the assumption of constant error variance (homoskedasticity). The test was significant, $\chi^2(1) = 212.91$, $p < .001$, indicating the presence of heteroskedasticity: the variance of the residuals is not constant across the range of fitted values. The Ramsey RESET test was used to check for omitted variables or model misspecification. This test was also significant, $F(3, 472) = 25.19$, $p < .001$, indicating that the model may be missing relevant predictors or a more complex functional form (for example, non-linear relationships between predictors such as Instagram followers and market value). A histogram of the model's residuals (Figure 1) was generated to visually assess whether the residuals were approximately normally distributed, a further assumption underlying the ordinary least squares estimates reported above.

Variable	**VIF**	**1/VIF**
No. of IG Followers	4.76	0.2099
Brand Industry Code	3.50	0.2860
No. of Matches Played	1.99	0.5034
Developing Country	1.68	0.5947
Age	1.01	0.9949
Mean VIF	2.59	

Discussion

Results of this research suggest that market value of a professional football player is influenced by both his experience on and off the field. As in previous studies that found a strong relationship between social media engagement and market value (Frenger et al., 2019; Cai, 2024), the Instagram follower count was a strong, significant, and positive predictor of market value, even when age, matches played, brand endorsement and nationality effects were accounted for. This strengthens the notion that clubs and the transfer market increasingly value an individual player's commercial reach and increased public profile alongside his/her on-field performance.

Concurrently, the number of matches played (which was also used as a proxy for experience and durability) was the strongest predictor in the model that has been previously found, that is experience-related variables are still very important for valuation (Margareta & Malinda, 2022). Interestingly, the study did not show a direct age effect on market value once the number of games played variable was taken into the equation, and this is slightly different to the findings of Sánchez et al. (2022) and Liu (n.d.). One likely possibility is that the interaction of age with value is mediated almost entirely by playing experience and reputation measured by number of matches played, as opposed to age per se itself being an independent factor once these other factors have been controlled for. The marginal effect for the brands endorsed by industries ($p = .054$) indicates a positive but less strong and less robust relationship than did the number of Instagram followers, which could be due to the smaller and less variable scale of this measure in this sample (mean = 1.01, SD = .11).

The coefficient for developing country was significant and positive, reflecting the fact that on average, this sample of players with developing countries had higher market values than players from developed

countries, after accounting for the other variables. With the coding of the variable (with values ranging from 0 to 3 instead of a standard binary 0/1), this result should be interpreted with caution because the variable may have more than just a developing/non-developing distinction, and it was not a core element of the original hypotheses.

These findings are qualified by the diagnostic tests. If there is heteroskedasticity, the standard errors reported for the coefficients can be biased and this will affect the magnitude of the significance test, especially on the borderline brand-industry coefficient. The significant Ramsey RESET result additionally suggests that the linear specification employed here is not entirely suitable to fully capture the association among the predictors and market value, for example, the relation between the number of Instagram followers and market value could be non-linear (logarithmic), as indicated in the extreme right-skew of the descriptive statistics (mean = 3,901 and maximum = 205,357).

Conclusion

The aim of this research was to investigate if there is a relationship between the value of professional football players on Instagram and their market value in general. The multiple regression model accounted for approximately 77% of the variance in market value, and revealed that the number of matches played, number of Instagram followers and developing-country origin were significant positive predictors, while brand-endorsement industries were significant, albeit somewhat less strongly, positive predictors, and age alone was not significant when these other factors were controlled for. The results indicate that off-field commercial visibility (namely the number of followers on social media) has become an important part of a player's market value, complementing and sometimes outcompeting other traditional market value-determining factors that rely on direct experience with the player, like age. For clubs, agents and sponsors, this is a sign of the increasing value of a player's digital image as an integral part of his or her commercial worth and not just a spin-off of a successful career.

Limitations and Future Research

These results need to be interpreted with certain caveats. First, the model revealed signs of heteroskedasticity and a large Ramsey RESET result, suggesting that some of the assumptions of OLS regression were not satisfied, so the constraints of this model may yield less robust estimates; a model with robust standard errors, a log transformed followee variable or a non-linear specification might be more useful. Second, the number of Instagram followers, brand-endorsement industries and matches played in the dataset was not obtained from a verified statistical database, but was produced by Transfermarkt which is an estimate and not a player’s actual transfer price. Third, this study is correlational, as opposed to experimental or longitudinal, and therefore, there is no conclusive evidence that social media following or brand endorsements increase the market value of a player it might be that the metrics of each one being already high will just accrue more. Prior literature has pointed to on-field performance measures (such as goals, assists, or advanced performance measures) and they were not directly added in this model, but may be added to future models to create a more comprehensive understanding of what factors influence market value.

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